WORKSHEET

Algebra and formulas

Part A

Use the values in the table below to determine the value of each expression.

Pronumeral	a	b	k	p	x	y
Value	2	10	20	4	3	12

1 $5a$

2 $b - a$

3 $\frac{k}{p}$

4 xy

5 $3x + b$

6 b^2

7 $2b + 3p$

8 $y \div p$

9 $a(k - 4x)$

10 $5p(y - 2)$

11 $\frac{ab}{k}$

12 $\frac{xy}{6}$

13 $p^2 + 2p$

14 $xy - ab$

15 $(a + 1) \times (p - 2)$

16 $(y - 7)^2$

17 $\frac{x + y}{5}$

18 $5b + k$

19 $5xp$

20 $\frac{k}{p} + \frac{y}{x}$

Part B

1 We can use the formula $D = \frac{8t}{25}$ to calculate the distance D km we are away from a storm when we hear the thunder t seconds after we see the lightning. Oscar counted 5 seconds between seeing lightning and hearing thunder. How far away is the storm?

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2 We can use Fried's formula to calculate a child's dose, D, of an adult's medicine. Fried's formula is $D = \frac{MA}{150}$. In the formula A = the adult dose and M = the child's age in months. The adult dose of cough medicine is 36 mL. How much cough medicine can these children have?

a Sally aged 28 months

b Cameron age 4 years

3 We can use the formula $P = 2(L + B)$ to calculate the perimeter P, of a rectangle. Calculate the perimeter of a rectangle with $L = 15$ and $B = 25$.

4 Fiona decided to make some of Grandma's biscuits. However the instructions said to set the oven to 400°F and the temperature gauge on Fiona's oven is in Celsius, °C. Use the formula $C = \frac{5}{9}(F - 32)$ to calculate the temperature in °C Fiona needs to cook Grandma's biscuits.

5 The currency in China is called Chinese Yuan Renminbi, or CYN. The formula to convert between CYN (C) and Australian dollars, (A), is $C = 5.19A$. Aisha is going to China for a holiday and she plans to take $650 for spending money. How many CYN will she get for $650 Australian?

6 Before refrigeration, it was difficult to keep milk fresh. We can calculate the number of days, D, milk will stay fresh at different temperatures t °C with the formula: $D = \frac{12}{t+1}$ where t is a number bigger than zero.

a How many days will milk stay fresh if it is stored at 5 °C?

b How many days longer will it stay fresh if it is stored at 3 °C compared to 5 °C?

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Answers

Part A

1 10
2 8
3 5
4 36
5 19
6 100
7 32
8 3
9 16
10 200
11 1
12 6
13 24
14 16
15 6
16 25
17 3
18 70
19 60
20 9

Part B

1 1.6 km
2 **a** 6.7 mL
b 11.5 mL
3 80
4 204°C
5 3373.5 CYN
6 **a** 2 days
b 1 more day